

📌 EVENT-DRIVEN CONGESTION · PLANNED & UNPLANNED

TRAVIS

Traffic Risk & Advisory Intelligence System

Forecasting event-driven traffic impact and recommending manpower, barricading & diversion plans for Bangalore.

TEAM BRIDGENSENSE

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LIVE PROTOTYPE

Deployed on Streamlit Cloud

3 ML models · Mappls

Events break the city — and we react too late

Rallies, festivals, sports events, construction and sudden gatherings create localized traffic breakdowns — but their impact is never quantified before they happen.

Why it's hard today



Not quantified

Event impact is never measured in advance — planners fly blind.



Experience-driven

Manpower & barricade decisions rely on individual gut-feel, not data.



No learning loop

Nothing is captured after an event to improve the next response.

The question we set out to answer

“How can historical and real-time data be used to forecast event-related traffic impact and recommend optimal manpower, barricading, and diversion plans?”

Our answer: TRAVIS

Turn any event into a quantified impact score, an explainable risk breakdown, and a ready-to-act deployment + diversion plan — in real time.

From a single event to a complete operating picture

1

PREDICT

3 ML models output closure risk, priority & duration → one 0–100 impact score.

2

EXPLAIN

SHAP shows which factors drove the risk; AI writes a plain-language advisory.

3

ACT

Deployment plan + live Mappls map with diversion routing & nearest units.

4

BROADCAST

One-click public bulletin authorities can post online for commuters.

Every step degrades gracefully — if a live API is down, TRAVIS falls back to model data and synthetic routing so it never breaks during operations.

Three models, one impact score

Road Closure

LightGBM classifier

→ **P(closure)**

AUC 0.72

Priority

LightGBM classifier

→ **P(high priority)**

AUC 0.99

Duration

LightGBM regressor

→ **resolution hours**

log-target

Impact Score = P(closure)·55 + cause severity·30 + duration·15

Engineered from 32 features — historical closure/priority rates per corridor, police station, junction & zone, cause severity rank, corridor fragility, and cyclical time encodings.

Tiering: **LOW < 25** **MEDIUM 25–49** **HIGH ≥ 50**

From prediction to operational decision



Resource Deployment Plan

Constables, barricade points & target response time, scaled to the risk tier.



Live Mappls Map

Lane-level data, real-time traffic layer, impact radius & nearest police stations.



Diversion Routing

Blocked road (red) vs diversion (green) with side-by-side ETAs — divert or stay.



SHAP Explainability

Top risk drivers shown per prediction — no black box.



AI Advisory (Groq)

3-sentence officer briefing generated strictly from model data.



Public Bulletin

One-click printable advisory for authorities to post online.

How the pieces fit together

FRONTEND

Streamlit web app (dark ops UI)
Interactive Mappls JS SDK map embed
Live on Streamlit Cloud

INTELLIGENCE

3 × LightGBM models (joblib)
32-feature engineering pipeline
SHAP attribution per prediction

INTEGRATIONS

Mappls: Map, Route, Distance Matrix,
Geocode, Traffic, Static Map
Groq llama-3.3-70b for NL advisory

Data flow

Event inputs → feature engineering → 3 models → impact score → Mappls map + deployment plan + diversion → SHAP + Groq advisory → public bulletin
Full architecture & flow diagrams included in the repo [\(Repo Link\)](#).

Explainable, grounded, and resilient

Explainable by design — Every prediction ships with a SHAP breakdown of its top risk drivers — officers see the 'why', not just the 'what'.

AI grounded in model data only — The Groq advisory receives only model outputs, SHAP drivers & nearest-station distances. It is instructed to invent nothing — no hallucinated streets, times or numbers.

Graceful degradation — If Mappls or Groq is unreachable, TRAVIS falls back to synthetic routing, straight-line distance and a deterministic template — it always produces a usable plan.

Real road intelligence — Distances and diversions use Mappls road-network APIs, not straight-line guesses — actionable for real deployment.

SEE IT LIVE

A working prototype, not a mockup

Try it in 30 seconds ▶ Open the live app

- ▶ **Pick a preloaded scenario** — e.g. Public Rally on Mysore Road
- ▶ Click Forecast Impact
- ▶ Read the impact score, map, diversion ETA & AI advisory
- ▶ Download the public traffic bulletin

LIVE APP

travis-traffic-risk-and-advisory-intelligence-system-6fy3zvxqh.streamlit.app

5 demo scenarios built in

Planned & unplanned event types

Source + diagrams in GitHub repo

THE IMPACT

Directly answers every gap in the problem statement

GAP TODAY

HOW TRAVIS CLOSSES IT

Impact not quantified

0–100 impact score for any event, before it happens

Experience-driven deployment

Data-driven manpower, barricade & response recommendations

No post-event learning

Model trained on historical events; retrainable as data grows

Plans hard to communicate

AI advisory + public bulletin make it readable for officers & citizens

Where TRAVIS goes next

Future Scope

- ▶ **Live event feeds** — auto-ingest from city calendars, police bulletins & social signals
- ▶ **Real-time multi-event city dashboard** — monitor every active disruption at once
- ▶ **Post-event learning loop** — log outcomes & continuously retrain (closes the feedback gap)
- ▶ **Citizen alerts** — push diversion advisories via app / SMS / Google Maps

Future Enhancements

- ▶ **Higher-accuracy models** — more data, deep temporal models, live traffic features
- ▶ Confidence bands on duration & impact predictions
- ▶ Auto-optimized diversion across multiple alternate routes
- ▶ Role-based access for control room, field officers & public portal
- ▶ Integration with VMS boards & signal control systems

Thank You

TRAVIS — Traffic Risk & Advisory Intelligence System

Quantify the event. Plan the response. Inform the city.

Team BridgeSense · Nirmal Trivedi · Darshan Bhavsar

Live: travis-traffic-risk-and-advisory-intelligence-system-6fy3zvxqh.streamlit.app

Github repo: <https://github.com/darshanbhavsar01/TRAVIS-Traffic-Risk-and-Advisory-Intelligence-System>

Demo Video: <https://drive.google.com/drive/folders/1jIA7U9v0KQtITb1xaUYEav873YX7OV0D?usp=sharing>